



OpenCV & UGOT Robot

Commands Cheat Sheet

Your Quick Reference Guide for Computer Vision & Robot Navigation

IMPORT STATEMENTS

These commands load the libraries we need:

Command	What It Does
<code>from ugot import ugot</code>	Loads the UGOT robot library
<code>import cv2</code>	Loads OpenCV for computer vision
<code>import numpy as np</code>	Loads NumPy for working with arrays/images
<code>import time</code>	Loads time module for delays
<code>from IPython.display import Image, display</code>	Loads display functions for Jupyter

OPENCV COMMANDS - Color Detection

These commands help the robot "see" and understand colors:

Command	What It Does
<code>cv2.cvtColor(img, cv2.COLOR_BGR2HSV)</code>	Converts image to HSV color space (easier to find colors!)
<code>cv2.inRange(hsv, lower, upper)</code>	Creates a mask showing only pixels within color range
<code>cv2.findContours(mask, cv2.RETR_EXTERNAL, cv2.CHAIN_APPROX_SIMPLE)</code>	Finds the outlines of colored objects
<code>cv2.boundingRect(contour)</code>	Gets x, y, width, height of a rectangle around the object

OPENCV COMMANDS - Drawing

These commands draw shapes and text on images:

Command	What It Does
<code>cv2.rectangle(img, (x1,y1), (x2,y2), color, thickness)</code>	Draws a rectangle (box) around objects
<code>cv2.circle(img, (cx,cy), radius, color, thickness)</code>	Draws a circle at a point (-1 = filled)
<code>cv2.line(img, (x1,y1), (x2,y2), color, thickness)</code>	Draws a straight line
<code>cv2.putText(img, text, (x,y), font, scale, color, thickness)</code>	Adds text message to the image

OPENCV COMMANDS - Image Processing

Command	What It Does
<code>cv2.imencode('.jpg', img)</code>	Converts image to JPEG format for display
<code>cv2.imdecode(nparr, cv2.IMREAD_COLOR)</code>	Converts camera data back to an image
<code>img.copy()</code>	Makes a copy of the image (protects original)

NUMPY COMMANDS

NumPy helps us work with image data as numbers:

Command	What It Does
<code>np.array([0, 100, 100])</code>	Creates an array (list of numbers) for HSV values
<code>np.zeros((height, width, 3), dtype=np.uint8)</code>	Creates a blank black image
<code>np.frombuffer(frame, np.uint8)</code>	Converts camera data to NumPy array

UGOT ROBOT COMMANDS

These commands control your UGOT robot:

Command	What It Does
<code>got.open_camera()</code>	Turns on the robot's camera
<code>got.read_camera_data()</code>	Takes a picture from the camera
<code>got.mecanum_move_xyz(x, y, z)</code>	Moves the robot: x=left/right, y=forward/back, z=rotation
<code>got.mecanum_stop()</code>	Stops all wheel movement

Movement Examples:

Command	Direction	Robot Movement
<code>mecanum_move_xyz(0, 10, 0)</code>	Forward	Robot moves straight ahead
<code>mecanum_move_xyz(5, 5, 0)</code>	Forward-Right	Robot moves diagonally right
<code>mecanum_move_xyz(-5, 5, 0)</code>	Forward-Left	Robot moves diagonally left
<code>mecanum_move_xyz(10, 0, 0)</code>	Strafe Right	Robot slides sideways right
<code>mecanum_move_xyz(0, 0, 10)</code>	Rotate Right	Robot spins clockwise

HSV COLOR REFERENCE

Common color ranges for detection (Hue, Saturation, Value):

Color	Lower Range	Upper Range
RED	[0, 100, 100]	[10, 255, 255]
GREEN	[35, 100, 100]	[85, 255, 255]
BLUE	[100, 100, 100]	[130, 255, 255]
YELLOW	[20, 100, 100]	[35, 255, 255]
ORANGE	[10, 100, 100]	[20, 255, 255]

QUICK TIPS

- Always use `img.copy()` before drawing on images to keep the original safe!
- Colors in OpenCV are BGR (Blue, Green, Red), not RGB!
- Use `time.sleep(0.1)` to prevent the loop from running too fast!
- The center of the camera image is usually around x=280 for a 560-wide image.
- Larger bounding box height = object is closer to the robot!

 **Happy Coding with OpenCV and UGOT Robots!** 